What Makes a Show Successful?

Streaming Content Analysis: Insights and Action Plan

1. Key Insights

Insight 1: Success distribution

- After removing 2 duplicate rows, the dataset contains 100 shows. A show is labelled successful when its viewership is at or above the median (51.0): 51 shows are successful and 49 are less successful (Figure 1).
- The two groups are almost evenly balanced, which makes the data well suited to a success/less-successful classification model.

Insight 2: Genre

- Average viewership: Sci-Fi 87.1, Action 74.7, Drama 58.6, Thriller 48.5, Documentary 40.2, Comedy 37.4 (Figure 2).
- Every Sci-Fi and Action show in the data is successful, compared with about 55% of Drama, 36% of Thriller, 10% of Documentary and 5% of Comedy shows.

Insight 3: Rating vs viewership

- Rating and viewership are only moderately related (correlation 0.57), so they do not always move together (Figure 3).
- Documentaries average a rating of 8.2, the same as Sci-Fi, yet their average viewership is only 40.2. Of the 40 shows rated 8.0 or higher, 12 are still less successful.

Insight 4: Language

- Average viewership by language: Korean 69.6, English 64.2, Japanese 55.5, Spanish 54.7, Hindi 46.4, French 35.4.
- Japanese (6 shows) and French (9 shows) have small samples, and language partly overlaps with genre mix, so these differences should be read with care.

Insight 5: Budget

- Budget has the strongest link to viewership (correlation 0.93). Shows with a budget of 80 or more average 80.9 viewership (36 shows), while shows under 40 average 36.0 (37 shows).
- Budget is also the top-ranked feature in the Random Forest (Figure 4). Because big-budget titles are concentrated in Sci-Fi and Action, this shows association, not proof that spending alone causes viewership.

Insight 6: Content combinations

- The strongest genre-language groups (3 or more shows) are Sci-Fi in English (94.2), Drama in Korean (85.3), Action in English (79.9) and Action in Spanish (77.2).

3. Visualisations

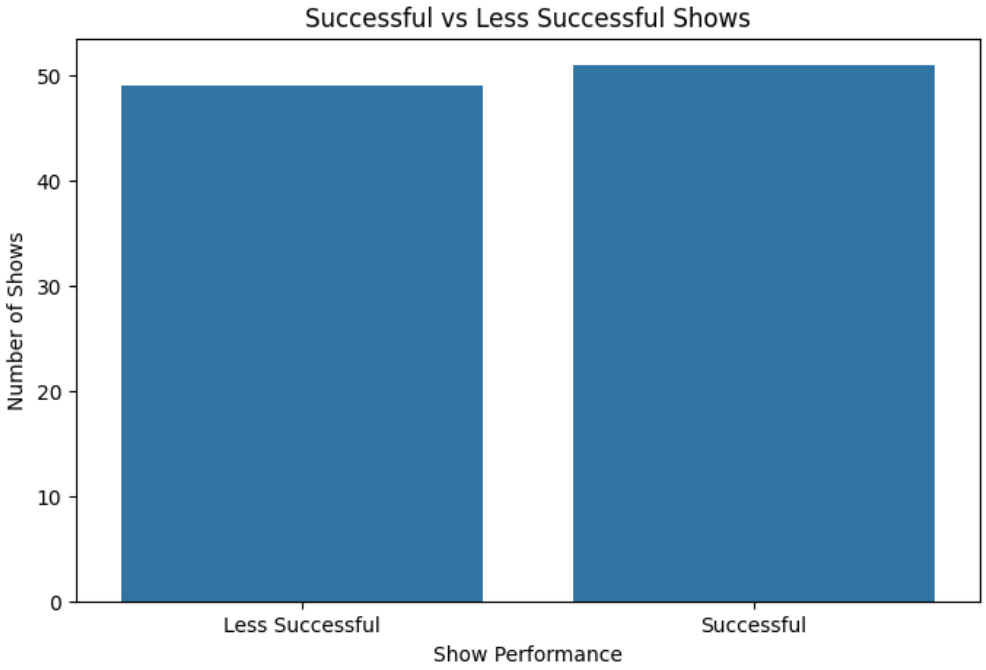


Figure 1: Successful vs less successful shows

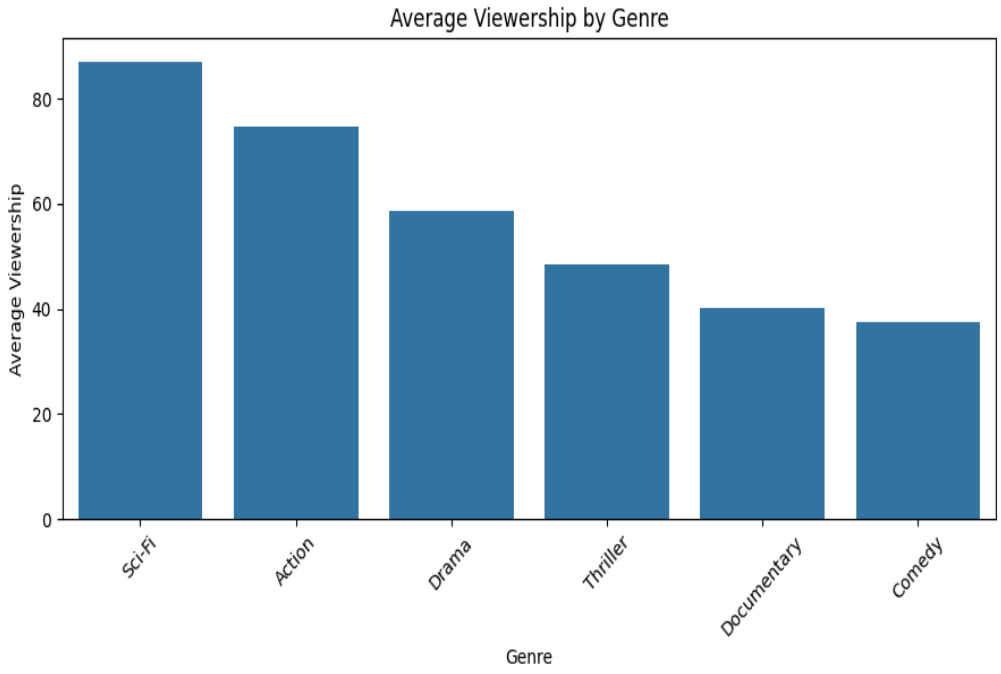


Figure 2: Average viewership by genre

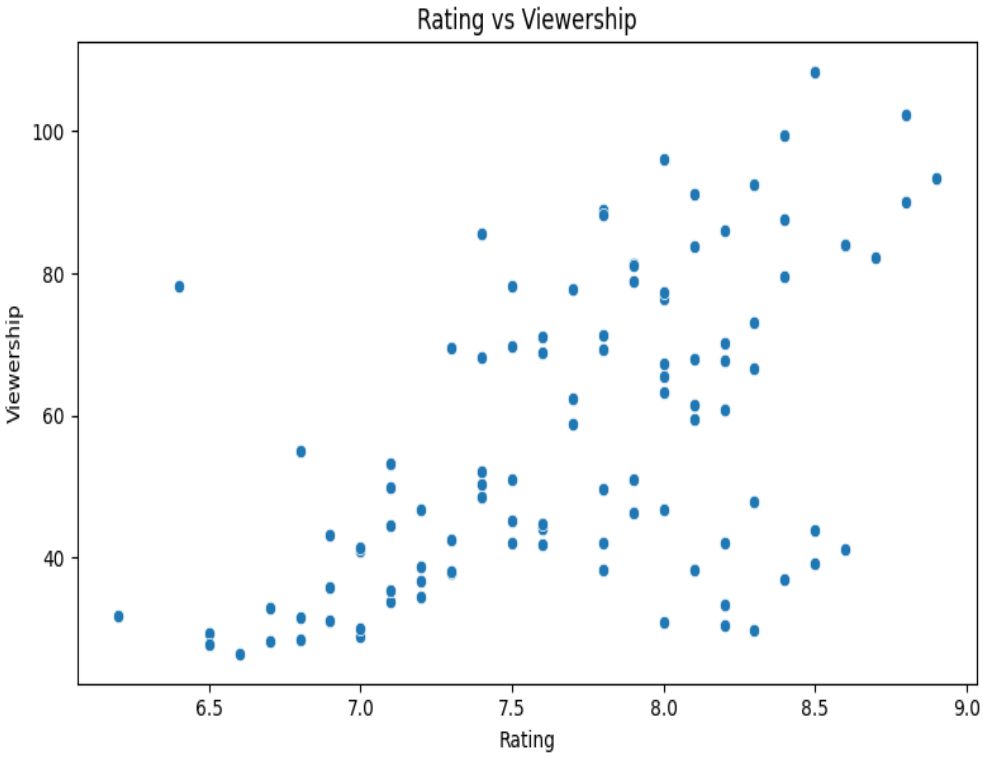


Figure 3: Rating vs viewership

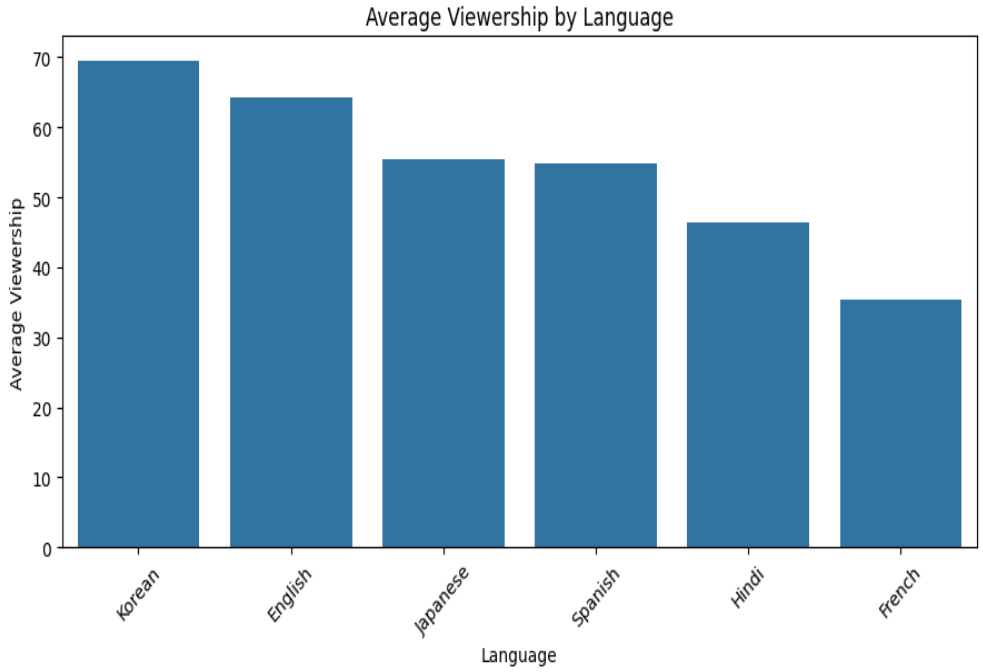


Figure 4: Top 10 Random Forest feature importances

4. Data Preparation Notes

- Removed 2 duplicate rows (SHOW_006 and SHOW_021).
- Standardised text values: "drama" to Drama, "ENGLISH" to English, "SciFi" to Sci-Fi.
- Treated invalid values as missing and filled them with the column median: negative runtime and budget, a rating of 15.5, and a viewership of 9999.
- Filled missing rating and budget with the median, and the missing genre with the most common genre.
- Defined success as viewership at or above the median.